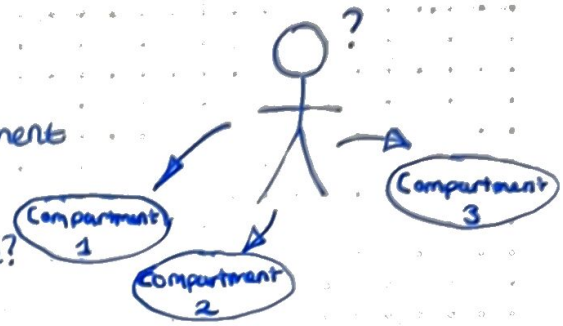
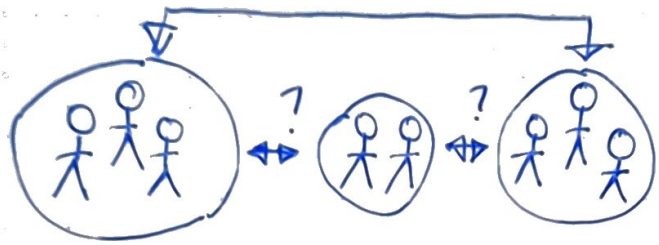
Building a Model

Compartmental Model

Any person classify to a particular compartment



How many people in compartment at given time?



Rules that govern rate of change!

Rate is amount per unit of time

$$\text{Rate} = \frac{\text{Quantity}}{\text{Time}}$$

Rate of change, how something changes relative to time

$$\text{Rate of Change} = \frac{\text{Change in Quantity}}{\text{Time of which change occured}}$$

Δ = delta is change in value

or written as

$$\frac{\Delta \text{ Quantity}}{\Delta \text{ Time}}$$

Building a mode Steps:

0 - Know your question
1 - Understand underlying processes
2 - Pick compartments → susceptible, infected, removed
3 - Make assumptions → contact rate? Probability of passing on the disease
4 - Draw flow diagram
5 - Write down model

SIR model

or SEIR
↳ exposed

Common Cold

1. Biology = everyone susceptible, mutates quickly - no immunity
2. Compartments = Susceptable, Infectious (no Removed)
3. Assumptions = Transmission Rate = β Recovery Rate = $\gamma = \frac{1}{\text{Duration of infection}}$ find in literature
4. Flow Diagram = $S \underset{\gamma I}{\overset{\frac{\beta \times I \times S}{N}}{\rightleftarrows}} I$
5. Model = $\Delta S = \gamma I - \frac{\beta I S}{N}$

$\Delta I = \frac{\beta I S}{N} - \gamma I$

Other compartments = A - asymptomatic
M - Maternal - immunity

↑ simplified - hasn't considered pre-symptomatic transmission for example

$\Delta R = \gamma I$ = Recovered

If $\Delta I > \gamma I$ then # infectious increasing

- Increasing $\frac{\beta S I}{N} > \gamma I$

$\frac{\beta S}{\gamma N} > 1$

one case means $I \neq 0$

Entirely susceptable pop $\frac{\beta}{\gamma} > 1$

Ebola

Incubation → 9 days
Infectious → 6 days

Dead & Infectious → 7 days
Dead & non-infectious → Removed

Fatality Rate → high

Immunity is long-term so unlikely to be reinfected

1 person infects = 0.5 to 2.5 people
R_0 estimate at 1.5